Nathan Zhang

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EDUCATION

The University of British Columbia Sep 2023 – May 2028
B.A., Combined Major in Economics and Statistics, Minor in Data Science **Average: 93.5/100**

- **Coursework:** Statistical Inference, Machine Learning and Data Mining, Causal Methods, Advanced Econometrics, Statistical Learning, Time Series and Forecasting, Computational Optimization
- **Awards:** Work Learn International Undergraduate Research Awards (2026), Economics Undergraduate Scholarship (2025), Faculty of Arts International Student Scholarship (2024, 2025)

EXPERIENCE

ESG Research Analyst Co-op Sep 2026 – Present
British Columbia Investment Management Corporation (BCI), Capital Markets & Credit Investments

- Apply causal machine learning to investigate how executive compensation and other governance risk factors affect financial performance, supporting risk-factor screening for fundamental investment research.
- Independently build reproducible Databricks ETL pipelines from Capital IQ and ISS ECA data, construct modeling variables, and develop ML models to predict M&A and other corporate events.
- Backtest voting strategies against historical votes with simulated ballot-time information, and analyze correlations with subsequent risk indicators to inform proxy voting.

Research Assistant Apr 2026 – Present
UBC Sauder School of Business

- Study technological innovation with PatentsBERTa patent and task embeddings, validated against Qwen 3 and manual reading. As sole RA for J. Frank Li and Cheng Chen, fit weighted FE OLS/GMM wage and sorting models, run large-scale estimation and bootstrap inference on HPC/Slurm, and draft the empirical section.
- Prepared financial alternative data from 3.2M+ ProQuest articles and video and audio metadata. Independently built reproducible ETL for RavenPack coverage checks, metadata-to-panel parsing, media archiving, deduplication, and subtitle validation (Allen Hu, May-Aug 2026).

Research Assistant Jan 2025 – May 2026
Vancouver School of Economics at UBC

- Developed CV, NLI, and sentiment-analysis teaching and research projects. Implemented image classification and human-validated zero-shot stance analysis, deployed interactive notebook and HTML courseware, and designed and taught a CV workshop.

PUBLICATIONS & MANUSCRIPTS

Sequential Tone Residualization and Abnormal Returns in Earnings Calls 2026
Joint first author, accepted at ICAIF 2026 and selected for oral presentation

- Proposed the earnings-call study and co-designed sequential information sets for tone residualization using financial controls and prior dialogue. Built the Capital IQ, IBES, and CRSP panel and proposed dictionary and LLM validation.
- Positive OOS R^2 for tone prediction on the 2025 holdout. Across 1,901 calls, answer residuals were positively associated with market-adjusted returns, while prespecified joint placebo tests were not significant.

Memory Response Paths and Path-Structured Context Compilation 2026
Joint first author, submitted to AAAI and under review

- Proposed Memory Response Paths to trace how memories affect LLM-agent decisions. Designed matched availability experiments and context-rendering ablations, built the 180-ecology benchmark and model API workflows, and conducted statistical analysis.
- A separate 144-item evaluation found PSCC improved LLM-judged quality by 6.9 percentage points over A-Mem with roughly one-third fewer API tokens.

TECHNICAL SKILLS

Programming and ETL: Python, R, SQL, PySpark, DuckDB, Polars, Databricks
Quantitative methods: panel econometrics, GMM, causal inference, time-series forecasting, machine learning
Text, image and multimodal: NLP, semantic embeddings, computer vision, multimodal ETL
Modeling and finance tools: PyTorch, scikit-learn, fixest, Bloomberg, BQL
Research workflows: Bash, HPC/Slurm, Git, agent-assisted coding with Codex and Claude Code